Positional Plagiocephaly and Cranial Orthotic Helmet Therapy

Randomized controlled trial evidence demonstrates that custom cranial orthotic helmets provide no added skull-rounding benefit over natural course for moderate-to-severe infant plagiocephaly by age 2.

TURAL-COURSE; LOW-FOR-SEVERE-SYNDROMIC-CRANIOSYNOSTOSIS | STATUS: DRAFT | DOMAIN: PHYSICAL-DEVELOPMENT, HEAL

WHAT WE KNOW (Narrowest Supported Conclusion)

Rigorous pragmatic randomized controlled trial evidence shows that custom cranial orthotic helmets (costing $2,000 to $3,500) provide no clinically meaningful or statistically significant advantage in skull shape recovery over the natural course (wait-and-see with physical repositioning and tummy time) by age 2 in infants with moderate-to-severe nonsynostotic positional plagiocephaly or brachycephaly. Spontaneous skull re-rounding occurs at statistically indistinguishable rates in both groups once infants achieve independent sitting and upright motor control. Prescribing a helmet for flat spot or for correcting plagiocephaly / brachycephaly routinely addresses parental cosmetic worry rather than underlying neurological risk.

EVIDENCE AND CAUSAL ASSESSMENT

- *van Wijk et al. (2014), BMJ** (The HEADS Study*, n=84): Pragmatic randomized controlled trial in the Netherlands enrolling healthy infants aged 5 to 6 months with moderate to severe positional skull deformation (plagiocephaly or brachycephaly). Infants were randomized to either 6 months of active helmet therapy (custom-molded passive orthosis worn 23 hours per day until age 12 months) or natural course without orthosis. The primary outcome was objective skull shape at 2 years of age measured by blinded plagiocephalometry (oblique diameter difference index [ODDI] for plagiocephaly and cranioproportional index [CPI] for brachycephaly).

- Primary RCT Findings: At 2 years of age, there was no significant difference in skull shape improvement between the helmet therapy group and the natural course group. The mean difference in ODDI was 0.2 (95% CI, -1.6 to 1.9, p=0.86), and the mean difference in CPI was 0.2 (95% CI, -1.7 to 2.1, p=0.80). Equal proportions of infants achieved full recovery: 25.6% in the helmet therapy group versus 22.5% in the natural course group (relative risk 1.14; 95% CI, 0.55 to 2.36).

- Adverse Effects & Burdens: In the helmet therapy arm, 100% of parents reported one or more adverse side effects during the 6-month wearing period: skin irritation/pressure sores (96%), augmented sweating (71%), unpleasant odor (76%), displacement of the helmet (60%), and skin pain/discomfort (33%).

EFFECT SIZE AND PERSISTENCE

The measured effect size of helmet therapy compared to natural course on objective cranial asymmetry at 2 years is essentially zero (standardized mean difference d ≈ 0.02). Spontaneous cranial remodeling is substantial across both arms between 6 and 24 months as the brain expands against gravity, skull bones thicken, and infants spend majority daytime hours sitting, crawling, and upright. Long-term follow-ups show no differences in cognitive, developmental, or motor outcomes between children treated with helmets versus those managed with observation.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (pragmatic randomized controlled trial testing a medical device intervention, not an observational heritability phenotype).

- Observational Confounding in Prior Commercial Literature: Non-randomized cohort studies showing "dramatic improvements" in helmet clinics compared treated infants at 6 months to their baseline at 4 months, confounding the helmet with the natural, spontaneous re-rounding that occurs in almost all non-syndromic infants as motor milestones emerge.

- Exclusion Boundaries: The trial explicitly excluded infants with true synostosis (premature cranial suture fusion requiring surgical repair), severe torticollis unresponsive to physical therapy, or dysmorphic syndromes.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

This evidence does not mean that positional flattening does not exist or that parents should avoid the supine "Back to Sleep" position for infant sleep. Back sleeping reduces Sudden Infant Death Syndrome (SIDS) by over 50% and is non-negotiable. Positional flattening is a benign cosmetic condition caused by prolonged external pressure on soft neonatal cranial bones; it does not compress the brain or impair neurocognitive development. Furthermore, this does not mean that premature fusion of cranial sutures (craniosynostosis) can be ignored; craniosynostosis is a structural pathological condition requiring pediatric surgical evaluation.

WHAT ACTION IS REASONABLY JUSTIFIED

Clinicians and parents can avoid expensive, uncomfortable cranial orthotic helmets ($2,000 to $3,500) for standard positional plagiocephaly and brachycephaly. The evidence strongly supports conservative, non-invasive management:

1. Maintain firm supine sleep for SIDS prevention.

2. Provide frequent, supervised daytime tummy time while awake from week one to relieve occipital pressure and strengthen neck and trunk musculature.

3. Alternate head position during naps and crib resting, and vary arm positions during feedings.

4. If congenital or positional muscular torticollis (restricted neck rotation) is present, initiate pediatric physical therapy early rather than relying on a passive helmet.

VERIFIED SOURCES & DOI IDENTIFIERS

- van Wijk, R. M., van Vlimmeren, L. A., Groothuis-Oudshoorn, C. G. M., Van der Ploeg, C. P. B., IJzerman, M. J., & Boere-Boonekamp, M. M. (2014). Helmet therapy in infants with positional skull deformation: randomised controlled trial. BMJ, 348, g2741. DOI: 10.1136/bmj.g2741 (https://doi.org/10.1136/bmj.g2741)